LIBBY — MANUSCRIPTS INVENTORY

Manuscripts considered — gist-sdh-multifocal-resected-m1-k4n8

Master inventory: every paper reviewed in this case

Generated 2026-05-19

Manuscripts considered — gist-sdh-multifocal-resected-m1-k4n8

Master inventory: 22 clinical (13 included, 9 considered & excluded) + 22 pre-clinical (14 included, 8 considered & excluded) + 11 additional trial publications/registrations. One entry per manuscript, sorted by year (newest first). Excluded entries are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

—/George (2026)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** pemigatinib PO daily on days 1-14 of a 21-day cycle
- **Indication / model:** advanced SDH-deficient GIST (2L+); histologically confirmed SDH-deficient GIST
- **Design:** open-label, single-arm phase 2
- **n:** 24
- **Effect size:** — — objective radiographic response rate (RECIST 1.1)
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Flynn/George (2025) — Cancer Research (AACR Annual Meeting Abstract CT215)

Reference:doi:10.1158/1538-7445.AM2025-CT215 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** pemigatinib / rogaratinib — FGFR inhibition in SDH-deficient GIST PDX
- **Indication / model:** patient-derived xenograft model of SDH-deficient GIST (PG-20), with parallel arms versus vehicle, sunitinib, regorafenib, rogaratinib, and pemigatinib
- **Design:** Genome-wide DNA hypermethylation in dSDH-GIST disrupts CTCF-bound chromatin insulators, leading to aberrant transcriptional activation of FGF3 and FGF4. Autocrine/paracrine FGF3/4 ligand signals through FGFR1 (and to a lesser extent FGFR4) to drive proliferation, so pharmacologic FGFR inhibition removes the downstream driver.
- **n:** PDX cohort 6-10 mice per arm (typical CT215 abstract footprint)
- **Effect size:** strong — Both rogaratinib and pemigatinib produced significant tumor growth suppression in the SDH-deficient GIST PDX, while sunitinib and regorafenib were essentially ineffective in the same model. Continuous FGFR-inhibitor exposure was required for sustained disease control. The clinical readout in the parallel rogaratinib arm of NCT04595747 (n=24) gave 41.7% ORR and 31-month median PFS.
- **Variance:** vs vehicle; sunitinib and regorafenib as TKI-of-record comparators; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** AACR meeting abstract, not full peer-reviewed paper. Single PDX model. The clinical signal (ORR 41.7%) is the strongest preclinical-to-clinic translation in dSDH-GIST so far and underwrites the new pemigatinib trial NCT07434843 the patient could enter at recurrence.

—/— (2025)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** trametinib +/- vebreltinib (MET inhibitor) on MAP2K1-alteration arm
- **Indication / model:** tumor-agnostic precision-medicine basket — MAP2K1-alteration arm (2L+); MAP2K1 activating alteration
- **Design:** open-label, multi-arm precision-medicine phase 2 basket
- **Effect size:** — — objective response rate (RECIST 1.1)
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Dedousis D/von Mehren M (2025) — Curr Treat Options Oncol

Reference:PMID 40045030 · **Type:** trial · **Inclusion:** included

- **Intervention:** regorafenib (3L wt-GIST anchor reference)
- **Indication / model:** SDH-deficient GIST systemic-therapy landscape (review) (3L+); SDH-deficient GIST
- **Design:** narrative review
- **Effect size:** regorafenib activity in dSDH-GIST: modest, similar to KIT-mutant 3L setting — —
- **Case fit:** partial
- **Notes:** (toxicity table not extracted)

Mavroeidis L/Jones RL (2025) — Expert Rev Gastroenterol Hepatol

Reference:PMID 40156874 · **Type:** trial · **Inclusion:** included

- **Intervention:** ripretinib (4L+ GIST anchor reference)
- **Indication / model:** rare GIST subtypes (SDH-deficient, NF1, BRAF-mutant) — review (4L+); SDH-deficient GIST
- **Design:** narrative review
- **Effect size:** ripretinib activity in dSDH-GIST: case-level only, no prospective wt-GIST cohort data — —
- **Case fit:** weak
- **Notes:** (toxicity table not extracted)

Jhaveri KL/Turner NC (2024) — New England Journal of Medicine

Reference:PMID 38446714 · **Type:** clinical · **Inclusion:** excluded — Same scope rule as SOLAR-1 — INAVO120 enrolled HR+/HER2- breast cancer with canonical PIK3CA hotspot mutations (E542K, E545K, H1047R, etc.). PFS HR 0.43 (p<0.0001) and OS 34.0 vs 27.0 mo (HR 0.67, p=0.019) are registration-grade, but the R93W non-canonical variant in this case has not been shown to confer comparable inavolisib sensitivity, and breast-cancer histology adds a second mismatch layer.

- **Intervention:** inavolisib + palbociclib + fulvestrant (INAVO120)
- **Notes:** Excluded: Same scope rule as SOLAR-1 — INAVO120 enrolled HR+/HER2- breast cancer with canonical PIK3CA hotspot mutations (E542K, E545K, H1047R, etc.). PFS HR 0.43 (p<0.0001) and OS 34.0 vs 27.0 mo (HR 0.67, p=0.019) are registration-grade, but the R93W non-canonical variant in this case has not been shown to confer comparable inavolisib sensitivity, and breast-cancer histology adds a second mismatch layer. — n=325. Captured for the audit trail so the board sees both PI3Kα-inhibitor pivotals were reviewed. The wt-GIST tersolisib trial (NCT05768139) the screener flagged is the right vehicle for any PI3Kα experiment, not the breast-cancer registration data.

—/— (2024)

Reference: — · **Type:** clinical · **Inclusion:** excluded — No published clinical readout for the wt-GIST cohort yet. LITESPARK-015 ASCO 2024/2025 data releases have covered the PPGL cohort (n=72, ORR 26%, DCR 85%, mPFS 22 mo) and the VHL/RCC cohorts, but the wt-GIST cohort has not reported. Trial-screener row already captures the protocol; no clinical-evidence row to write until the readout appears.

- **Intervention:** belzutifan (LITESPARK-015 wt-GIST cohort)
- **Notes:** Excluded: No published clinical readout for the wt-GIST cohort yet. LITESPARK-015 ASCO 2024/2025 data releases have covered the PPGL cohort (n=72, ORR 26%, DCR 85%, mPFS 22 mo) and the VHL/RCC cohorts, but the wt-GIST cohort has not reported. Trial-screener row already captures the protocol; no clinical-evidence row to write until the readout appears. — Audited so the board knows the dossier reviewed the belzutifan wt-GIST evidence base and confirmed it is still pre-readout. Watch for ASCO 2026 / WCGIC for the GIST-cohort interim cut.

Flego/Casali (2024) — Journal of Experimental & Clinical Cancer Research

Reference:PMID 38132569 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** temozolomide / MGMT methylation predictor in dSDH-GIST
- **Indication / model:** human GIST tumor tissue methylation profiling: SDH-deficient WT GIST vs KIT-mutant GIST vs SDH-proficient WT GIST, MGMT promoter methylation by pyrosequencing
- **Design:** Succinate accumulation from SDH loss inhibits TET-mediated DNA demethylation, with preferential drift toward CpG island hypermethylation. The MGMT promoter is one of the CpG islands silenced in this background, which lowers the threshold for temozolomide-induced cytotoxicity.
- **n:** n=24 KIT/PDGFRA-WT GIST (mix of dSDH and SDH-proficient) vs n=44 KIT-mutant comparator
- **Effect size:** moderate — MGMT promoter hypermethylation was preferentially enriched in dSDH-GIST relative to KIT-mutant and SDH-proficient WT-GIST controls. This is the molecular biomarker that explains why dSDH-GIST might respond to temozolomide where most other GIST molecular subtypes do not, and it justifies running MGMT methylation as a companion test at recurrence.
- **Variance:** vs KIT-mutant GIST and SDH-proficient WT GIST; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Descriptive correlative paper, not a drug study. Connects the upstream SDH biology to the downstream alkylator vulnerability that Yebra 2022 demonstrated functionally. Together they give the full mechanism stack for the temozolomide rationale.

Ligon JA/Glod J (2023) — Clinical Cancer Research

Reference:PMID 36302175 · **Type:** clinical · **Inclusion:** included

- **Intervention:** guadecitabine (SGI-110)
- **Indication / model:** SDH-deficient GIST, SDH-deficient PPGL, and HLRCC-associated RCC (2L+); 9 patients enrolled before trial closed for low accrual (target 70); age 18-57, 6F/3M; 7 SDH-deficient GIST, 1 PPGL, 1 HLRCC-RCC. Trial used a Simon two-stage design (target ORR 30%, rule out 5%).
- **Design:** open-label, single-arm phase 2 (terminated for low accrual)
- **n:** 9
- **Effect size:** 0 % — objective response rate (RECIST 1.1)

- **Toxicities:** neutropenia (treatment-limiting) G>=3 (2/9, 22.2%)
- **Case fit:** partial
- **Notes:** Negative phase 2 in the exact mechanism-matched population (DNA hypomethylating agent for dSDH-driven hypermethylation phenotype). Closes the hypomethylating-agent strategy for this patient at recurrence. Per-AE detail beyond the headline neutropenia signal not in the abstract.

Singh C/Pashankar F (2023) — Pediatric Blood & Cancer

Reference:PMID 36151992 · **Type:** clinical · **Inclusion:** included

- **Intervention:** olaparib + temozolomide (PARP inhibitor + alkylator combination)
- **Indication / model:** metastatic, multiply-relapsed SDH-deficient GIST + paraganglioma (single patient) (5L+); Single pediatric/young-adult patient with SDHB-mutant GIST plus paraganglioma; multiply relapsed after prior TKI lines. Yale single-institution case report.
- **Design:** single-patient case report
- **n:** 1
- **Effect size:** clinical response reported — clinical response (case-level)
- **Toxicities:** Case-report; per-AE detail not in the published letter.
- **Case fit:** weak
- **Notes:** Letter-format case report with no abstract. The only published clinical evidence for PARP + alkylator in dSDH-GIST. Kept for the recurrence-contingency mechanism context — not enough signal to act on alone.

Ligon/Glod (2023) — Clinical Cancer Research

Reference:PMID 36302175 · **Type:** preclinical · **Inclusion:** excluded — This is the phase II clinical trial result (0/9 responses), already captured in clinical_evidence.jsonl. Not preclinical. The mechanism-vs-clinic disconnect (genome-wide hypermethylation in tumor, but no objective responses to a global hypomethylator) is captured here as audit context: the Killian 2013 / Killian 2014 mechanism papers (kept) explain why the drug was tested, the Ligon trial explains why it is no longer rationally pursued.

- **Intervention:** guadecitabine in dSDH-GIST/PPGL/HLRCC RCC (Ligon 2023)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: This is the phase II clinical trial result (0/9 responses), already captured in clinical_evidence.jsonl. Not preclinical. The mechanism-vs-clinic disconnect (genome-wide hypermethylation in tumor, but no objective responses to a global hypomethylator) is captured here as audit context: the Killian 2013 / Killian 2014 mechanism papers (kept) explain why the drug was tested, the Ligon trial explains why it is no longer rationally pursued.

Gao/Solit (2023) — Molecular Cancer Therapeutics

Reference:PMID 36442478 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** trametinib / MEK inhibitors — MAP2K1 P124 functional characterization (Gao 2018)
- **Indication / model:** Ba/F3 and NIH3T3 IL-3-independent growth and transformation assays across a panel of MAP2K1 variants (including P124L, P124Q, P124S); BRAF V600E melanoma cell lines for the co-mutation arm
- **Design:** MAP2K1 P124 sits in the negative-regulatory alpha-C helix region. P124S, P124L, and P124Q substitutions destabilize the inactive conformation and allow constitutive RAF-dependent phosphorylation of MEK1, with downstream ERK activation. The variants remain RAF-dependent — they are class I activators, not

RAF-independent.
- **n:** n=100 MAP2K1 variants surveyed; replicate Ba/F3 and 3T3 assays per variant
- **Effect size:** moderate — P124L, P124Q, and P124S variants drove Ba/F3 IL-3 independence and NIH3T3 transformation. The variants retained sensitivity to trametinib and selumetinib at clinically achievable concentrations, classifying them as RAF-dependent (class I) activators where MEK monotherapy should hit. Helps the dossier separate P124 from RAF-independent (class II/III) MAP2K1 variants that are MEK-refractory.
- **Variance:** vs MAP2K1-WT and known activating variants (E203K, K57N) as positive controls; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Functional-variant survey, not a tumor-specific PDX. The classification of P124S as RAF-dependent and MEK-inhibitor-sensitive is the relevant cross-tumor signal; whether the variant is clonal in this patient's tumor remains a tissue-confirmation question. The variant is ctDNA-only at present.

Singh/Pashankar (2023) — Pediatric Blood & Cancer

Reference:PMID 36151992 · **Type:** preclinical · **Inclusion:** excluded — Single-patient case report, already captured in clinical_evidence.jsonl. Not preclinical. The Sulkowski 2018 and Sulkowski 2020 rows (kept) carry the mechanism this case rests on.

- **Intervention:** olaparib + temozolomide single-patient case (Singh 2023)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Single-patient case report, already captured in clinical_evidence.jsonl. Not preclinical. The Sulkowski 2018 and Sulkowski 2020 rows (kept) carry the mechanism this case rests on.

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** temozolomide 200 mg/m^2 PO daily x 5 days every 28 days
- **Indication / model:** advanced SDH-deficient GIST (2L+); KIT/PDGFRA-WT GIST with SDH deficiency (SDHB-IHC loss or SDHx mutation); exploratory cohort of WT-GIST without SDH deficiency
- **Design:** open-label, two-cohort, single-arm phase 2
- **Effect size:** — — objective response rate in the SDH-deficient WT-GIST cohort (RECIST 1.1)
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** tersolisib (STX-478) PO daily (dose-escalation)
- **Indication / model:** PIK3CA-mutant advanced solid tumors and HR+/HER2- breast cancer (2L+); PIK3CA activating mutation
- **Design:** first-in-human dose-escalation + expansion, monotherapy and combination
- **Effect size:** — — safety, MTD, pharmacokinetics, preliminary ORR
- **Case fit:** weak

- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Yebra M/Sicklick JK (2022) — Clinical Cancer Research

Reference:PMID 34426440 · **Type:** clinical · **Inclusion:** included

- **Intervention:** temozolomide (in SDH-mutant GIST)
- **Indication / model:** advanced SDH-mutant GIST (SDHA/SDHB/SDHC/SDHD mutation, biallelic in tumor) (2L+); 5 SDH-mutant GIST patients (1 SDHA, 2 SDHB, 1 SDHC, 1 SDHD); median age 29.7; 2F/3M; median 1 prior TKI line (range 0-2). Sicklick / UCSD case series paired with patient-derived xenograft modeling.
- **Design:** translational case series — patient-derived SDH-mutant GIST models with parallel clinical-cohort treatment-response capture
- **n:** 5
- **Effect size:** 40 % — objective response rate to TMZ in SDH-mutant GIST
- **Toxicities:** Paper does not publish a per-AE safety table for the clinical cohort; the translational focus is on the PDX-driven mechanism rationale. Standard TMZ low-dose continuous schedule toxicity profile (cytopenias, fatigue, nausea) presumed.
- **Case fit:** strong
- **Notes:** The case-series readout that motivated NCT03556384 (UCSD) and NCT05661643 (Asan). PMID 35443107 in the user prompt was a typo — the correct PMID is 34426440. Toxicities[] empty because the clinical-cohort section of the paper reports outcomes but not per-AE tables; PMC full text confirmed.

Giger OT/Casey RT (2022) — Journal of Clinical Pathology

Reference:PMID 36198483 · **Type:** clinical · **Inclusion:** included

- **Intervention:** MGMT promoter methylation as TMZ-response biomarker in SDH-deficient GIST
- **Indication / model:** SDH-deficient and KIT/PDGFRA-WT GIST (any); 65 GIST tumors profiled: 47 wt-GIST (33 SDH-deficient), 21 tyrosine-kinase-mutant. Single-institution Cambridge cohort.
- **Design:** retrospective molecular cohort with MGMT-methylation profiling
- **n:** 65
- **Effect size:** MGMT hypermethylation in 8/33 (24%) SDH-deficient GIST; 0% in TKI-mutant or SDH-preserved wt-GIST % — frequency of MGMT promoter hypermethylation by GIST subtype
- **Case fit:** strong
- **Notes:** The published biomarker rationale for testing MGMT methylation if temozolomide enters the discussion at recurrence. The 24% MGMT-methylated subset is the patient population most likely to respond to TMZ in the dSDH-GIST setting. Aligns with the target_validation file's MGMT methylation row. — (toxicity table not extracted)

Bayley/Devilee (2022) — Endocrine-Related Cancer

Reference:PMID 35324464 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** belzutifan / PT2977 in SDHB-deficient pheochromocytoma
- **Indication / model:** in vivo allograft of spontaneously immortalized mouse chromaffin (MPC) cells with CRISPR Sdhb knockout, plus matched parental Sdhb-WT MPC cells
- **Design:** Sdhb loss stabilizes HIF-2alpha through succinate accumulation. Belzutifan blocks HIF-2alpha:ARNT

heterodimerization and shuts off the HIF-2 transcriptional output (VEGFA, CXCR4, EPO) that drives the pseudohypoxia program.

- **n:** mouse cohorts 6-10 per arm (typical Endocrine-Related Cancer figure size)
- **Effect size:** moderate — Belzutifan reduced tumor growth in the Sdhb-knockout allograft model relative to vehicle, with parallel suppression of HIF-2 target genes. The effect was the most consistent of the targeted agents tested in this paired comparison, though absolute tumor regression was partial rather than complete.
- **Variance:** vs vehicle and parallel arms with mitochondrial complex I inhibitors, multikinase TKIs, and PARP inhibitors; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** SDHB-deficient pheochromocytoma model, not GIST, but the upstream lesion (SDH loss -> succinate -> HIF-2) is the same biology this patient carries. Allograft is murine, not patient-derived, so absolute response magnitudes should not be over-read. Closest published in vivo readout for belzutifan in an SDH-loss context — there is no published belzutifan study in an SDH-deficient GIST PDX yet.

Yebra/Sicklick (2022) — Clinical Cancer Research

Reference:PMID 34426440 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** temozolomide in SDH-deficient GIST — Yebra preclinical PDX establishment
- **Indication / model:** patient-derived 2D and 3D scaffold-free GIST cultures from freshly procured SDH-mutant GIST tissue (4 dSDH-GIST patient lines), with parallel KIT-mutant GIST line comparator
- **Design:** MGMT promoter hypermethylation (a downstream consequence of the SDH-loss hypermethylator phenotype) silences O-6-methylguanine DNA methyltransferase. Without MGMT to strip the temozolomide-induced O6-methylguanine adduct, mismatch-repair attempts the strand cycle, double-strand breaks accumulate, and the cell triggers apoptosis.
- **n:** n=4 patient-derived dSDH-GIST models; n=3 biological replicates per drug condition
- **Effect size:** strong — Patient-derived dSDH-GIST models recapitulated the transcriptional and metabolic hallmarks of the parent tumors and were sensitive to temozolomide via on-target DNA damage and apoptosis, while the imatinib/sunitinib/regorafenib arms produced no growth inhibition. The companion clinical mini-cohort (n=5 patients) gave a 40% ORR and 100% disease control rate to TMZ.
- **Variance:** vs DMSO vehicle; KIT-mutant GIST line as drug-control comparator; multikinase TKI arms (imatinib, sunitinib, regorafenib) in parallel; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** The decisive mechanism paper for the temozolomide trials this patient could enter at recurrence (NCT03556384, NCT05661643). Patient-derived models, on-target damage, paired clinical readout — about as good as it gets for a rare-tumor preclinical-to-clinical bridge. The 4-line model panel is small but the effect was consistent across all lines.

—/— (2021)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** belzutifan 120 mg PO daily
- **Indication / model:** advanced wild-type (KIT/PDGFRA-WT) GIST; also PPGL, pNET, VHL-associated tumors, HIF-2α-altered solid tumors (2L+); KIT/PDGFRA wild-type GIST (the dSDH cohort is the canonical wt-GIST subset)

- **Design:** open-label, multi-cohort, single-arm phase 2
- **Effect size:** — — objective response rate (BICR per RECIST 1.1)
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2021)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** DFF332 PO daily
- **Indication / model:** advanced/relapsed renal cancer and other HIF-stabilizing malignancies (SDHx-mutated PGL/PPGL expansion arm planned) (2L+); HIF-stabilizing alterations including SDHA/B/C/D/AF2 mutations
- **Design:** open-label dose-escalation +/- combinations (everolimus, IO)
- **n:** 40
- **Effect size:** — — safety, MTD, pharmacokinetics
- **Case fit:** none
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

von Mehren M/Van den Abbeele AD (2020) — Clinical Cancer Research

Reference:PMID 31792037 · **Type:** clinical · **Inclusion:** included

- **Intervention:** linsitinib (OSI-906, IGF-1R/IR inhibitor)
- **Indication / model:** KIT/PDGFRA wild-type GIST (adult and pediatric) (2L+); 20 patients with WT-GIST enrolled in stage 1 of the SARC phase II trial. Rationale anchored on IGF1R overexpression observed preferentially in dSDH-GIST.
- **Design:** SARC open-label, single-arm phase 2
- **n:** 20
- **Effect size:** 0 % — RECIST 1.1 ORR at 6 months; clinical benefit rate at 9 months
- **Toxicities:** grade 3-4 treatment-related AEs (composite) G>=3 (1.8%); nausea (4.9%); fatigue (3.2%); abnormal liver function (3.2%); diarrhea (2.5%)
- **Case fit:** partial
- **Notes:** Hard negative on ORR. The 40% clinical-benefit rate at 9 months overlaps the natural-history indolence of dSDH-GIST, so it is not informative for drug activity. Closes IGF-1R inhibition as a strategy in this molecular subset. Reported AE rates are expressed as percent of drug-related events, not patient-level percentages — flagged in toxicities[] notes.

Joensuu H/Reichardt P (2020) — JAMA Oncology

Reference:PMID 32469385 · **Type:** clinical · **Inclusion:** excluded — Same mechanism-scope rule — 10-year follow-up of the SSGXVIII trial population (KIT-mutant high-risk GIST, not SDH-deficient). 10-yr OS 79.0% (36-mo arm) vs 65.3% (12-mo arm); HR 0.55 (95% CI 0.37-0.83; p=0.004) reinforces the long-term benefit of extended adjuvant imatinib in the indications represented in the trial, but the dSDH-GIST subset is structurally absent.

- **Intervention:** adjuvant imatinib 36 vs 12 months — 10-year follow-up (SSGXVIII)
- **Notes:** Excluded: Same mechanism-scope rule — 10-year follow-up of the SSGXVIII trial population

(KIT-mutant high-risk GIST, not SDH-deficient). 10-yr OS 79.0% (36-mo arm) vs 65.3% (12-mo arm); HR 0.55 (95% CI 0.37-0.83; p=0.004) reinforces the long-term benefit of extended adjuvant imatinib in the indications represented in the trial, but the dSDH-GIST subset is structurally absent. — Long-term confirmation that the SSGXVIII benefit holds out to a decade — captured here so the audit trail shows the adjuvant-imatinib evidence was reviewed in full, not dismissed on the original 2012 paper alone.

Blay JY/von Mehren M (2020) — Lancet Oncology

Reference:PMID 32511981 · **Type:** clinical · **Inclusion:** included

- **Intervention:** ripretinib (INVICTUS pivotal RCT)
- **Indication / model:** advanced GIST after >=3 prior TKIs (imatinib, sunitinib, regorafenib) (4L+); 129 patients (85 ripretinib, 44 placebo) with >=3 prior TKIs. KIT-mutant disease dominated; prospective dSDH-GIST stratification not performed. 29 hospitals in 12 countries.
- **Design:** randomized double-blind placebo-controlled phase 3 (2:1)
- **n:** 129
- **Effect size:** 0.15 HR — progression-free survival (BICR)
- **Variance:** 95% CI 0.09–0.25; p=<0.0001
- **Toxicities:** lipase increase G>=3 (n=85, 5%); hypertension G>=3 (n=85, 4%); fatigue G>=3 (n=85, 2%); alopecia; hand-foot skin reaction
- **Case fit:** weak
- **Notes:** INVICTUS pivotal for 4L+ GIST. The Mavroeidis 2025 review notes the dSDH-GIST evidence base for ripretinib is the weakest of the three post-imatinib TKIs — INVICTUS enrolled vanishingly few wt-GIST patients. Captured as 4L contingency anchor.

—/— (2020)

Reference: — · **Type:** clinical · **Inclusion:** excluded — Standard POLE → ICI rationale does not apply to this case. The published POLE-hypermutator → ICI-response literature (Rousseau 2021 J Clin Oncol; Domingo 2016; Bourdais 2017) is anchored on the canonical exonuclease/proofreading-domain hotspots P286R, V411L, S459F, S297F — variants that produce TMB ~100+ mut/Mb and ultramutated phenotypes. R1679 sits outside the exonuclease domain and the patient is TMB <1 mut/Mb and MSS. The variant is most plausibly a VUS with no ICI rationale; cancer-predisposition counseling is the right action.

- **Intervention:** ICI in POLE proofreading-deficient hypermutators (rationale check)
- **Notes:** Excluded: Standard POLE → ICI rationale does not apply to this case. The published POLE-hypermutator → ICI-response literature (Rousseau 2021 J Clin Oncol; Domingo 2016; Bourdais 2017) is anchored on the canonical exonuclease/proofreading-domain hotspots P286R, V411L, S459F, S297F — variants that produce TMB ~100+ mut/Mb and ultramutated phenotypes. R1679 sits outside the exonuclease domain and the patient is TMB <1 mut/Mb and MSS. The variant is most plausibly a VUS with no ICI rationale; cancer-predisposition counseling is the right action. — Captured to document that the dossier reviewed the POLE → ICI literature and the standard rationale does not transport. No separate per-paper rows for the POLE-hypermutator pembrolizumab data because the threshold question (canonical hotspot, TMB-high, MSI-H) is failed at the genotype level. Aligns with the validation file's POLE-counseling row.

Sulkowski/Bindra (2020) — Nature

Reference:PMID 32494005 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** olaparib + temozolomide / oncometabolite-chromatin axis — Sulkowski 2020 Nature
- **Indication / model:** human cell lines with conditional IDH1/2-mutant, FH-knockdown, or SDHB-knockdown backgrounds; ChIP for H3K9me3 at DSB sites; pharmacologic and genetic KDM4B rescue
- **Design:** Succinate, fumarate, and 2-hydroxyglutarate all inhibit KDM4B at the chromatin around DNA double-strand breaks. The resulting local H3K9me3 hypermethylation prevents TIP60-mediated H4K16 acetylation, blocks 53BP1 release and BRCA1 recruitment, and shuts down HR at the break site even when the HR machinery itself is intact.
- **n:** n=3 biological replicates per condition
- **Effect size:** strong — KDM4B over-expression in succinate- or fumarate-elevated cells restored HR function and rescued PARP-inhibitor sensitivity, confirming KDM4B as the bottleneck. This locked in the mechanistic claim that the BRCA-like phenotype in SDH/FH/IDH-mutant tumors lives at chromatin, not at the HR protein level.
- **Variance:** vs isogenic WT cells; KDM4B over-expression rescue arm; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Mechanism-resolution sequel to Sulkowski 2018. Confirms the HR-defect logic at chromatin level but does not add new in vivo SDH-GIST evidence. Justifies pairing a PARP inhibitor with an alkylator (TMZ) in the SDH-deficient setting at recurrence — the Singh 2023 case report (PMID 36151992) is the only human readout so far.

—/— (2020)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** cobimetinib 60 mg PO daily, days 1-21 of a 28-day cycle
- **Indication / model:** tumor-agnostic, off-label molecularly-matched basket — NF1 or MAP2K1 alteration cohort (2L+); NF1 LOF or MAP2K1 activating mutation
- **Design:** MEGALiT — molecularly-guided off-label phase 2 basket
- **Effect size:** — — disease control rate per RECIST 1.1
- **Case fit:** weak
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

André F/Juric D (2019) — New England Journal of Medicine

Reference:PMID 31091374 · **Type:** clinical · **Inclusion:** excluded — Cross-tumor extrapolation candidate but the PIK3CA R93W variant in this case sits outside the canonical alpelisib companion-diagnostic hotspots (E542K, E545K/A/D/G, H1047R/Y/L, C420R). SOLAR-1 enrolled HR+/HER2- breast cancer with the validated hotspot variants; the registration data (mPFS 11.0 vs 5.7 mo; HR 0.65, p<0.001) cannot be transported to a non-canonical PIK3CA variant in a histology where PI3K-pathway activation has no established driver role.

- **Intervention:** alpelisib + fulvestrant (SOLAR-1)
- **Notes:** Excluded: Cross-tumor extrapolation candidate but the PIK3CA R93W variant in this case sits outside the canonical alpelisib companion-diagnostic hotspots (E542K, E545K/A/D/G, H1047R/Y/L, C420R). SOLAR-1 enrolled HR+/HER2- breast cancer with the validated hotspot variants; the registration data (mPFS 11.0 vs 5.7 mo; HR 0.65, p<0.001) cannot be transported to a non-canonical PIK3CA variant in a histology where PI3K-pathway activation has no established driver role. — n=572 (PIK3CA-mutant cohort). The dossier needs the explicit exclusion call because PIK3CA-mutant is one of the patient's user-stated targetable features. The

right path forward is tissue confirmation of R93W and a sponsor-by-sponsor variant-eligibility check rather than borrowing the SOLAR-1 effect size.

Wehn/Wallace (2019) — Journal of Medicinal Chemistry

Reference:PMID 31062976 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** belzutifan / PT2385 medicinal chemistry
- **Indication / model:** biochemical and biophysical HIF-2alpha PAS-B binding assays, crystallography, and in vivo VHL-null ccRCC xenograft (786-O) pharmacodynamic studies
- **Design:** PT2385 occupies a hydrophobic cavity inside the HIF-2alpha PAS-B domain, preventing the conformational rearrangement required for HIF-2alpha to dimerize with ARNT. The drug is selective for HIF-2alpha over HIF-1alpha — the PAS-B pocket of HIF-1alpha is too small to accept the ligand.
- **n:** n not standardized; multiple cell-line and PD experiments
- **Effect size:** moderate — PT2385 dissociated HIF-2 complexes in cells, suppressed HIF-2 target-gene transcription, and produced tumor growth inhibition in 786-O xenografts with acceptable PK. The medicinal chemistry from PT2385 led directly to PT2977 (belzutifan), which is ~10-fold more potent in xenograft models with cleaner human PK.
- **Variance:** vs vehicle; HIF-1alpha control to test selectivity; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** VHL-null ccRCC model again, not SDH-deficient GIST. Kept for the drug-mechanism story (what binds where, why selectivity exists) that justifies the belzutifan label expansion the patient might use at recurrence. Wallace 2016 (Cancer Research, PMID 27679553) is the discovery paper for PT2385; the 2019 Wehn/Wallace JMC is the PT2977 (belzutifan) optimization paper.

Raut CP/DeMatteo RP (2018) — JAMA Oncology

Reference:PMID 30383140 · **Type:** clinical · **Inclusion:** excluded — Same scope rule: PERSIST-5 enrolled intermediate/high-risk primary GIST without SDH-status stratification. 5-year RFS 90% and OS 95%, but recurrences clustered in patients with imatinib-insensitive genotypes (PDGFRA D842V, KIT-WT) — and 49% of the cohort stopped treatment early. The PI3K/AKT-driven recurrence pattern after discontinuation reinforces that imatinib is suppressing imatinib-sensitive clones, not eradicating disease. dSDH-GIST is the canonical imatinib-insensitive subset.

- **Intervention:** adjuvant imatinib 5 years (PERSIST-5)
- **Notes:** Excluded: Same scope rule: PERSIST-5 enrolled intermediate/high-risk primary GIST without SDH-status stratification. 5-year RFS 90% and OS 95%, but recurrences clustered in patients with imatinib-insensitive genotypes (PDGFRA D842V, KIT-WT) — and 49% of the cohort stopped treatment early. The PI3K/AKT-driven recurrence pattern after discontinuation reinforces that imatinib is suppressing imatinib-sensitive clones, not eradicating disease. dSDH-GIST is the canonical imatinib-insensitive subset. — n=91. The PERSIST-5 design itself is the cleanest illustration of why adjuvant imatinib doesn't apply here: the trial separated out the imatinib-insensitive genotypes and showed they recur on or off treatment. dSDH-GIST belongs to the same insensitive class.

Mei L/Boikos SA (2018) — Trends in Cancer

Reference:PMID 29413424 · **Type:** clinical · **Inclusion:** included

- **Intervention:** natural history and management of SDH-deficient GIST (narrative review)
- **Indication / model:** SDH-deficient and KIT/PDGFRA-WT GIST (any); Narrative review covering dSDH-GIST natural history, surveillance, surgical recurrence patterns, and TKI-response data from the NIH wt-GIST clinic and published cohorts.
- **Design:** narrative review (precision-medicine framing)
- **Effect size:** median EFS ~2.5 yr post-resection in the NIH 76-patient surgical cohort; recurrence or progression in ~71% of dSDH-GIST patients years — synthesis of dSDH-GIST recurrence rates, sites, and TKI activity by subtype
- **Case fit:** strong
- **Notes:** The reference review the board will reach for to ground the surveillance question. Recurrence is the rule in dSDH-GIST even after R0 resection — drives the case for biennial whole-body MRI and PGL screening rather than adjuvant systemic therapy. — (toxicity table not extracted)

Gounder MM/Tap WD (2018) — New England Journal of Medicine

Reference:PMID 26222557 · **Type:** clinical · **Inclusion:** included

- **Intervention:** trametinib (in MAP2K1-mutant histiocytic sarcoma)
- **Indication / model:** histiocytic sarcoma with MAP2K1 F53L activating mutation (2L+); Single adult patient with metastatic histiocytic sarcoma, MAP2K1 F53L (negative regulatory domain), prior chemotherapy. MSKCC.
- **Design:** single-patient case report (NEJM letter)
- **n:** 1
- **Effect size:** complete response — complete clinical and radiographic response
- **Toxicities:** Single-patient report; trametinib tolerated at standard 2 mg daily dose without dose modification reported.
- **Case fit:** cross_tumor_only
- **Notes:** Cross-tumor precedent for MEK monotherapy in a MAP2K1-activating-mutation context, but F53L is in the negative-regulatory domain (class I MAP2K1 mutation) and this patient's P124S is at the αC-helix interface (class II behavior in melanoma data). Cross-tumor + cross-class — informs mechanism plausibility but does not transport effect size.

Sulkowski/Bindra (2018) — Nature Genetics

Reference:PMID 30013182 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** olaparib + temozolomide / PARP-trapping in SDH-loss models — Sulkowski 2018 Nat Genet
- **Indication / model:** SDHB-knockdown and FH-knockdown human cell lines (HeLa, U2OS) with HR-repair functional readouts (RAD51 foci, DR-GFP); in vivo nude-mouse xenograft tumor-growth-delay assay with PARP inhibitor BMN-673 (talazoparib)
- **Design:** Succinate and fumarate competitively inhibit the alpha-ketoglutarate-dependent KDM4A/B histone lysine demethylases. The resulting aberrant H3K9 trimethylation at DSB sites occludes recruitment of the homologous-recombination machinery (53BP1, BRCA1, RAD51). With HR locked out, cells must rely on more error-prone repair, and PARP-trapping inhibitors push them past the survival threshold — a BRCA-like synthetic-lethal state with no actual BRCA mutation.
- **n:** in vitro n=3 biological replicates; in vivo n=8-10 mice per arm (typical for Nat Genet figures of this kind)
- **Effect size:** strong — SDHB or FH knockdown produced a measurable HR defect (loss of RAD51 foci, fall in DR-GFP repair signal) and dramatically sensitized cells to olaparib and talazoparib. The xenograft arm showed significant tumor growth inhibition with talazoparib in SDH-/FH-deficient tumors relative to isogenic controls.

- **Variance:** vs isogenic SDHB-WT or FH-WT cells; vehicle in vivo; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** SDHB/FH knockdown cells rather than authentic dSDH-GIST PDX, so the molecular phenotype is right but the tumor histology is not matched. Most directly relevant preclinical anchor for the Singh 2023 PARP+TMZ case report (PMID 36151992) and the rationale for a contingency PARP regimen at recurrence.

—/Burgoyne (2018)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** temozolomide 85 mg/m^2 PO daily x 21 days every 28 days
- **Indication / model:** advanced SDH-mutant/deficient GIST (2L+); SDH-mutant or SDH-deficient GIST (SDHA / SDHB / SDHC / SDHD mutation or SDHB-IHC loss)
- **Design:** open-label, single-arm phase 2
- **Effect size:** — — overall response rate at 6 months (RECIST 1.1)
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

—/— (2018)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** no investigational drug; clinical evaluation, imaging, germline + tumor genomics
- **Indication / model:** rare solid tumors (SDH-deficient GIST explicitly on the eligible list) (n/a); SDH-deficient GIST or other rare-tumor / tumor-predisposition phenotype
- **Design:** longitudinal natural-history observational registry with biospecimen acquisition
- **n:** 10000
- **Effect size:** — — natural-history characterization; biospecimen banking
- **Case fit:** strong
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted; primary endpoint not extracted)

Andritsos LA/Grever MR (2017) — Leukemia & Lymphoma

Reference:PMID 28918710 · **Type:** clinical · **Inclusion:** excluded — Cross-tumor and cross-class. MAP2K1 K57N is a class I activating mutation in the negative-regulatory domain — biochemically distinct from the αC-helix P124 hotspot in this case. The hematologic-malignancy histology adds a second layer of mismatch. Single-patient hairy cell leukemia case with cutaneous response on trametinib does not transport to a wt-GIST patient.

- **Intervention:** trametinib (in MAP2K1 K57N variant hairy cell leukemia)
- **Notes:** Excluded: Cross-tumor and cross-class. MAP2K1 K57N is a class I activating mutation in the negative-regulatory domain — biochemically distinct from the αC-helix P124 hotspot in this case. The hematologic-malignancy histology adds a second layer of mismatch. Single-patient hairy cell leukemia case with cutaneous response on trametinib does not transport to a wt-GIST patient. — Reviewed to anchor whether any class-matched MEK-inhibitor data exists for the patient's P124S. They do not — the closest published precedents are class-I MAP2K1 mutations in non-GIST histologies.

Boikos SA/Helman LJ (2016) — JAMA Oncology

Reference:PMID 27011036 · **Type:** clinical · **Inclusion:** included

- **Intervention:** sunitinib (in KIT/PDGFRA-WT, SDH-deficient GIST)
- **Indication / model:** KIT/PDGFRA wild-type GIST (NIH Pediatric & Wild-Type GIST Clinic cohort; majority SDH-deficient) (2L+); 95 patients with adequate tumor specimens from the NIH wt-GIST clinic; median age 23; 70% female; molecular subtypes: SDH-mutant (n=63), SDH-epimutant (n=21), SDH-competent (n=11). The cohort skews young and pediatric-onset, with gastric primaries dominating the SDH-deficient strata.
- **Design:** observational natural-history cohort with molecular subtyping plus retrospective treatment-response capture
- **n:** 95
- **Effect size:** imatinib 1/49 (2%) PR in SDH-deficient; sunitinib 7/38 (18%) objective response in same group % — objective response stratified by molecular subtype and drug
- **Toxicities:** Toxicity not the focus of this cohort report; sunitinib and imatinib safety as expected for label dosing.
- **Case fit:** partial
- **Notes:** The landmark molecular-subtype reference from the NIH Pediatric & wt-GIST Clinic. Treatment-response data are retrospective (not from a prospective sub-trial) and the per-drug n's are small, but they remain the most-cited published anchor for sequencing TKIs in dSDH-GIST. Toxicity capture not feasible — paper is a registry/cohort analysis without per-AE tables.

Chen/Wallace (2016) — Nature

Reference:PMID 27595394 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** belzutifan / PT2977 (HIF-2alpha inhibitor) — Chen PT2399 preclinical proof
- **Indication / model:** VHL-mutant clear cell renal cell carcinoma patient-derived xenografts in mice (n=18 distinct PDX models), plus VHL-null RCC cell lines
- **Design:** PT2399 binds the lipophilic PAS-B pocket of HIF-2alpha and blocks heterodimerization with ARNT (HIF-1beta), so HIF-2alpha cannot drive transcription of VEGFA, EPO, CCND1 and the rest of the HIF-2 program — even when HIF-2alpha protein is stabilized.
- **n:** n=18 PDX models, group sizes typical 5-10 mice/arm
- **Effect size:** moderate — PT2399 dissociated HIF-2 in VHL-null RCC cells and suppressed tumor growth in 10 of 18 (56%) PDX models, with activity exceeding sunitinib in head-to-head arms and durable disease control in a sunitinib-progressing tumorgraft. The sensitive vs resistant split was tracked to an HIF-2-dependent gene signature, foreshadowing the patient-selection problem.
- **Variance:** vs vehicle and sunitinib comparator arm; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** VHL-null ccRCC, not SDH-deficient GIST. The link to this patient is mechanistic: VHL loss and SDH loss both converge on stabilized HIF-2alpha. Cross-tumor extrapolation, not direct model match. Cho 2016 (PMID 27728802, Nature) is the companion paper that replicated the result in an independent PDX series.

Cho/Kaelin (2016) — Nature

Reference:PMID 27728802 · **Type:** preclinical · **Inclusion:** excluded — Companion paper to Chen 2016, same conclusion (PT2399 regresses VHL-null ccRCC xenografts) on the Kaelin lab cohort. Chen 2016 already represents the proof-of-concept in the dossier; including both would be double-counting.

- **Intervention:** belzutifan / PT2399 — Cho 2016 companion paper
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Companion paper to Chen 2016, same conclusion (PT2399 regresses VHL-null ccRCC xenografts) on the Kaelin lab cohort. Chen 2016 already represents the proof-of-concept in the dossier; including both would be double-counting.

Wagle N/Garraway LA (2014) — Cancer Discovery

Reference:PMID 24265154 · **Type:** clinical · **Inclusion:** included

- **Intervention:** MAP2K1 P124 hotspot characterization (resistance mechanism in BRAF-mutant melanoma on RAF/MEK combo)
- **Indication / model:** BRAF-mutant melanoma with acquired resistance to dabrafenib + trametinib (any); Five melanoma patients with progression on combined RAF + MEK inhibition; tumor biopsies sequenced at progression.
- **Design:** molecular cohort — paired biopsies at baseline and progression
- **n:** 5
- **Effect size:** MEK2 Q60P identified as the dominant on-treatment resistance variant; MAP2K1 P124 mutations characterized as preexisting low-frequency alterations that diminish but do not eliminate BRAF-inhibitor response — MAPK-pathway alterations at progression on RAF/MEK combination
- **Case fit:** cross_tumor_only
- **Notes:** The reference paper for the framing of MAP2K1 P124 as a partial-resistance / partial-activating αC-helix variant. In melanoma, P124L/S/Q diminish dabrafenib response but tumors retain MEK-inhibitor sensitivity in vitro — the cleanest published mechanism rationale for considering trametinib in this patient's P124S, while flagging that the response data is melanoma-only and the variant alone has not anchored a positive-readout trial. — (toxicity table not extracted)

Killian/Meltzer (2014) — Science Translational Medicine

Reference:PMID 25540324 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** guadecitabine / decitabine — SDHC promoter epimutation rationale
- **Indication / model:** human SDHx-WT GIST tissue (n=16) profiled by methylation array and bisulfite sequencing at the SDHC promoter; paired with SDHB IHC and SDH-complex enzymology
- **Design:** Hypermethylation of the SDHC promoter silences SDHC transcription in tumors that lack a coding SDHx mutation. This is itself a maintained methylation defect, which closes the loop: SDH loss begets hypermethylation begets SDHC silencing begets more SDH loss.
- **n:** n=16 SDHx-WT dSDH GIST cases
- **Effect size:** strong — SDHC promoter CpG island hypermethylation and gene silencing was present in 15 of 16 (94%) SDHx-WT dSDH GIST cases. The finding unified the genetic and epigenetic ends of dSDH-GIST pathogenesis under one mechanism and gave a direct substrate for hypomethylating-agent rescue.
- **Variance:** vs SDHx-mutant dSDH GIST and KIT-mutant GIST; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** The patient's tumor is SDHA-mutant rather than SDHC-epimutant, so the SDHC-specific mechanism applies less directly. The broader hypermethylator phenotype still holds. Note that the clinical trial that came out of this mechanism (Ligon 2023 guadecitabine, PMID 36302175) returned 0/9 responses — the mechanism-vs-clinic disconnect for this patient is real.

Wagle/Garraway (2014) — Cancer Discovery

Reference:PMID 25370473 · **Type:** preclinical · **Inclusion:** excluded — Wagle 2014 framed MAP2K1 P124 in the BRAF-inhibitor resistance setting in melanoma. Relevant for the BRAF-mutant context but the patient is BRAF-WT, so the resistance angle does not load here. The Gao 2018 panel (kept) is the better-fit functional reference for monotherapy-MEK consideration in a non-BRAF context.

- **Intervention:** MAP2K1 P124 — BRAF-inhibitor resistance context (Wagle 2014)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Wagle 2014 framed MAP2K1 P124 in the BRAF-inhibitor resistance setting in melanoma. Relevant for the BRAF-mutant context but the patient is BRAF-WT, so the resistance angle does not load here. The Gao 2018 panel (kept) is the better-fit functional reference for monotherapy-MEK consideration in a non-BRAF context.

Demetri GD/Casali PG (2013) — Lancet

Reference:PMID 23177515 · **Type:** clinical · **Inclusion:** included

- **Intervention:** regorafenib (GRID pivotal RCT)
- **Indication / model:** advanced GIST after imatinib + sunitinib failure (3L+); 199 patients (133 regorafenib, 66 placebo) with progression after both imatinib and sunitinib. No prospective SDH-status stratification; KIT-mutant disease dominated the population.
- **Design:** randomized double-blind placebo-controlled phase 3 (2:1)
- **n:** 199
- **Effect size:** 0.27 HR — progression-free survival (BICR)
- **Variance:** 95% CI 0.19–0.39; p=<0.0001
- **Toxicities:** hypertension G>=3 (31/132, 23%); hand-foot skin reaction G>=3 (26/132, 20%); diarrhea G>=3 (7/132, 5%)
- **Case fit:** partial
- **Notes:** GRID pivotal — the registration-grade evidence base for the 3L TKI in the contingency plan. dSDH-GIST under-represented in the cohort; activity in this subset is by extrapolation plus the Dedousis 2025 review's qualitative claim of 'modest activity, similar to KIT-mutant 3L'.

Killian/Meltzer (2013) — Cancer Discovery

Reference:PMID 23550148 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** guadecitabine / decitabine (DNA-hypomethylating agents) — SDH-deficient GIST methylator biology
- **Indication / model:** human GIST tumor samples profiled by Illumina Infinium 450K methylation array; SDH-deficient (n=12) vs KIT-mutant (n=23) cohorts with paired normal tissue
- **Design:** Succinate inhibits the alpha-ketoglutarate-dependent TET 5-methylcytosine dioxygenases that drive active DNA demethylation. The downstream consequence in GIST is genome-wide CpG hypermethylation that silences tumor-suppressor and differentiation programs.
- **n:** n=12 SDH-deficient GIST, n=23 KIT-mutant GIST, plus normal tissue references
- **Effect size:** strong — SDH-deficient GIST carried roughly 10-fold more hypermethylated CpG targets than KIT-mutant GIST (84,900 vs 8,400 sites). The signature was global and quantitative, not driven by a handful of loci, and it tracked SDHB-IHC loss across the cohort. This established the GIST methylator phenotype that

mechanistically justifies hypomethylating-agent trials.

- **Variance:** vs KIT-mutant GIST and adjacent normal gastric tissue; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Patient-tissue methylome paper, not a drug study. Sets the rationale for guadecitabine but cannot itself predict response.

Belalcazar/Trent (2013) — Molecular Diagnosis & Therapy

Reference:PMID 23046288 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** linsitinib / IGF1R — IGF1R overexpression and inhibition in WT-GIST (preclinical)
- **Indication / model:** GIST cell lines (including KIT-mutant GIST-T1 and GIST-882) with IGF1R siRNA knockdown and IGF1R small-molecule inhibitors; expression-survey work on KIT/PDGFRA-WT human GIST tissue
- **Design:** SDH loss stabilizes HIF-1alpha, which transcriptionally upregulates IGF1R. The receptor is overexpressed at mRNA and protein level in dSDH-GIST without gene-level amplification, and the active receptor signals through PI3K-AKT and MAPK to drive proliferation independent of KIT.
- **n:** n=3 biological replicates per cell-line condition
- **Effect size:** moderate — IGF1R knockdown and IGF1R small-molecule inhibition induced apoptosis in GIST cell lines and arrested proliferation. The dSDH-GIST tissue surveys confirmed IGF1R overexpression in 70-90% of cases. This was the preclinical engine for the SARC phase II linsitinib trial — which then returned 0/20 objective responses in the clinic.
- **Variance:** vs scrambled siRNA; DMSO vehicle; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** partial
- **Notes:** Preclinical work was done largely on KIT-mutant GIST cell lines because dSDH-GIST cell lines did not exist at the time. The IGF1R-overexpression observation is real, but the clinical linsitinib trial (von Mehren 2020, PMID 31792037) is a hard negative for ORR. Kept as audit-trail evidence so the board sees the IGF1R hypothesis was tested and clinically discredited.

Lasota/Miettinen (2013) — American Journal of Surgical Pathology

Reference:PMID 23232851 · **Type:** preclinical · **Inclusion:** excluded — Large IHC correlative survey establishing 88% IGF1R overexpression in dSDH-GIST. Descriptive rather than functional/mechanistic; the Belalcazar 2013 row already carries the functional IGF1R-knockdown evidence. Audit row kept to flag that the IHC base for the IGF1R hypothesis is documented even if the linsitinib trial flopped.

- **Intervention:** IGF1R IHC in 1078 GIST (Lasota)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Large IHC correlative survey establishing 88% IGF1R overexpression in dSDH-GIST. Descriptive rather than functional/mechanistic; the Belalcazar 2013 row already carries the functional IGF1R-knockdown evidence. Audit row kept to flag that the IHC base for the IGF1R hypothesis is documented even if the linsitinib trial flopped.

Joensuu H/Reichardt P (2012) — JAMA

Reference:PMID 22453568 · **Type:** clinical · **Inclusion:** excluded — Same population mismatch as Z9001:

SSGXVIII enrolled high-risk localized GIST by modified NIH consensus criteria with no SDH-status stratification, and the trial pre-dates the routine identification of SDH-deficient GIST. The 36-month vs 12-month finding (RFS HR 0.46; OS HR 0.45) is the basis of the 3-year adjuvant standard in KIT-mutant high-risk disease but does not transport to dSDH-GIST. ESMO-EURACAN-GENTURIS recommends against adjuvant imatinib in SDH-deficient GIST.

- **Intervention:** adjuvant imatinib 36 vs 12 months (SSGXVIII/AIO)
- **Notes:** Excluded: Same population mismatch as Z9001: SSGXVIII enrolled high-risk localized GIST by modified NIH consensus criteria with no SDH-status stratification, and the trial pre-dates the routine identification of SDH-deficient GIST. The 36-month vs 12-month finding (RFS HR 0.46; OS HR 0.45) is the basis of the 3-year adjuvant standard in KIT-mutant high-risk disease but does not transport to dSDH-GIST. ESMO-EURACAN-GENTURIS recommends against adjuvant imatinib in SDH-deficient GIST. — n=397. Cited frequently in adjuvant-GIST discussions; the dossier needs the explicit exclusion call because the surgical team may ask 'why aren't we doing 3 years of adjuvant imatinib?' — answer: because the patient is in a subset that was not represented and that does not respond.

Burke/Williams (2012) — PNAS

Reference:PMID 22949673 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** alpelisib / PI3Kalpha inhibitors — PIK3CA ABD-domain mutation biology
- **Indication / model:** purified recombinant PI3Kalpha (p110alpha/p85) lipid kinase assays with engineered ABD-domain variants (R88Q, G106V, and related linker mutations); hydrogen-deuterium exchange mass spectrometry to map conformational changes
- **Design:** The ABD (adaptor binding domain, residues ~1-108) packs against the kinase domain and helps clamp p110alpha in an inactive state. Mutations at conserved residues in this interface — R88, R93, G106 — disrupt the ABD/kinase-domain hydrogen bond network and let the kinase domain swing into its active conformation, raising basal lipid kinase activity without ligand input.
- **n:** n=3 biochemical replicates per variant
- **Effect size:** weak — ABD-domain mutations (R88Q, G106V and related) elevated basal lipid kinase activity above WT but to a markedly lesser degree than the canonical helical-domain (E545K) and kinase-domain (H1047R) hotspots. The 'gain of function' is real but partial — these are weakly activating variants, not strong oncogenic drivers, and the response profile to PI3Kalpha inhibitors is correspondingly less predictable.
- **Variance:** vs PIK3CA-WT enzyme; hotspot variants (E545K, H1047R) as positive controls; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** weak
- **Notes:** Biochemistry paper, no tumor model. R93W specifically is not individually profiled in this dataset — extrapolation runs through R88Q and the surrounding ABD residues. Supports the validator's caution that R93W is non-canonical and clinical actionability is uncertain. Companion Burke/Williams 2013 paper (PMID 23818588) extended the structural model to the helical and kinase hotspots.

Janeway KA/Helman LJ (2011) — PNAS

Reference:PMID 21173220 · **Type:** clinical · **Inclusion:** included

- **Intervention:** SDH-deficient GIST molecular subtype identification
- **Indication / model:** KIT/PDGFRA wild-type GIST (any); 34 patients with KIT/PDGFRA-WT GIST without personal or family history of paraganglioma; tested for germline SDHB/C/D mutations.

- **Design:** prospective molecular cohort (Pediatric GIST Clinic, NIH/DFCI)
- **n:** 34
- **Effect size:** 4/34 (12%) WT-GIST patients carried germline SDH mutations % — frequency of germline SDH defects in wt-GIST
- **Case fit:** strong
- **Notes:** The paper that established SDH-deficient GIST as a real molecular entity beyond the Carney-Stratakis syndrome. Frames the diagnostic logic the board will use to interpret the SDHA L306P + Y408N + E350fs constellation in this case. — (toxicity table not extracted)

DeMatteo RP/Owzar K (2009) — Lancet

Reference:PMID 19303137 · **Type:** clinical · **Inclusion:** excluded — Trial enrolled localized primary KIT-positive GIST (CD117+) without molecular stratification; SDH-deficient GIST was not a recognized subtype in 2009 and the dSDH-GIST population was not represented in the cohort. The ESMO-EURACAN-GENTURIS guideline explicitly recommends avoiding adjuvant imatinib in SDH-deficient GIST because imatinib has near-zero response in this molecular subset (Boikos 2016: 1/49 imatinib PR). The Z9001 1-year imatinib vs placebo result (RFS HR 0.35, 98% vs 83% at 1 yr; p<0.0001) does not transport to this patient.

- **Intervention:** adjuvant imatinib (ACOSOG Z9001)
- **Notes:** Excluded: Trial enrolled localized primary KIT-positive GIST (CD117+) without molecular stratification; SDH-deficient GIST was not a recognized subtype in 2009 and the dSDH-GIST population was not represented in the cohort. The ESMO-EURACAN-GENTURIS guideline explicitly recommends avoiding adjuvant imatinib in SDH-deficient GIST because imatinib has near-zero response in this molecular subset (Boikos 2016: 1/49 imatinib PR). The Z9001 1-year imatinib vs placebo result (RFS HR 0.35, 98% vs 83% at 1 yr; p<0.0001) does not transport to this patient. — Captured for the audit trail as the foundational adjuvant-imatinib RCT (n=713). The board needs to see that the absence-of-evidence in dSDH-GIST is structural — these patients were never enrolled in the registration trials.

Pasini/Stratakis (2008) — European Journal of Human Genetics

Reference:PMID 18301448 · **Type:** preclinical · **Inclusion:** excluded — Genetic-syndrome description of Carney-Stratakis (SDHB/C/D germline GIST + paraganglioma) rather than primary preclinical drug data. Belalcazar 2013 (kept) covers the IGF1R-overexpression mechanism with experimental data.

- **Intervention:** IGF1R in Carney-Stratakis / dSDH-GIST (Pasini)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Genetic-syndrome description of Carney-Stratakis (SDHB/C/D germline GIST + paraganglioma) rather than primary preclinical drug data. Belalcazar 2013 (kept) covers the IGF1R-overexpression mechanism with experimental data.

Demetri GD/Casali PG (2006) — Lancet

Reference:PMID 17046465 · **Type:** clinical · **Inclusion:** included

- **Intervention:** sunitinib (post-imatinib pivotal RCT)
- **Indication / model:** advanced/metastatic GIST after imatinib failure or intolerance (2L+); 312 patients (207 sunitinib, 105 placebo) with imatinib-resistant or -intolerant GIST. No SDH-status stratification; the cohort was overwhelmingly KIT-mutant by historical molecular epidemiology, with dSDH-GIST under-represented.

- **Design:** randomized double-blind placebo-controlled phase 3 (2:1)
- **n:** 312
- **Effect size:** 0.33 HR — time to progression
- **Variance:** p=<0.0001
- **Toxicities:** fatigue; diarrhea; skin discoloration; nausea
- **Case fit:** partial
- **Notes:** Pivotal RCT supporting sunitinib's FDA approval for 2L GIST. Captured as 'included' because (a) Boikos 2016 establishes sunitinib is the preferred 2L TKI in dSDH-GIST and (b) this is the registration-grade safety + efficacy anchor for the contingency plan at first recurrence. Detailed per-AE table beyond the abstract not retrieved here.

King/Gottlieb (2006) — Oncogene

Reference:PMID 16892081 · **Type:** preclinical · **Inclusion:** excluded — Narrative review, not primary preclinical data. Selak 2005 (kept) is the source experimental work; the Killian 2013 / Killian 2014 papers cover the GIST-specific extension.

- **Intervention:** belzutifan / SDH-loss biology (King review)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Narrative review, not primary preclinical data. Selak 2005 (kept) is the source experimental work; the Killian 2013 / Killian 2014 papers cover the GIST-specific extension.

Selak/Gottlieb (2005) — Cancer Cell

Reference:PMID 15652751 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** belzutifan (HIF-2alpha inhibitor) / SDH-loss pseudohypoxia biology
- **Indication / model:** in vitro biochemistry with purified HIF prolyl hydroxylase (PHD2) plus succinate-loaded mammalian cell extracts; SDH-knockdown cells in normoxia
- **Design:** Succinate that accumulates when SDH activity drops escapes the mitochondrion into the cytosol and competitively inhibits the alpha-ketoglutarate-dependent prolyl hydroxylases (PHD1-3) that normally tag HIF-1alpha for VHL-mediated degradation. The result is HIF-1alpha stabilization under normal oxygen tension, the so-called pseudohypoxia phenotype.
- **n:** n=3 biological replicates per condition (typical for the figures)
- **Effect size:** strong — SDH inhibition raised intracellular succinate to mM concentrations, blocked PHD activity in vitro at physiologically relevant succinate levels, and produced HIF-1alpha protein accumulation in normoxic cells. Adding cell-permeable alpha-ketoglutarate reversed the effect, locking in the mechanism as competitive inhibition at the 2OG-binding pocket of PHD.
- **Variance:** vs untreated cells in normoxia; alpha-ketoglutarate add-back; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Mechanism paper, not a therapeutic study. This is the foundational biology that places HIF-2alpha (and DNA/histone demethylases) downstream of SDH loss in this patient. King 2006 (Oncogene PMID 16892081) is the paired review that generalized the model to SDH-mutant paraganglioma.

Hegi/Stupp (2005) — New England Journal of Medicine

Reference:PMID 15758010 · **Type:** preclinical · **Inclusion:** excluded — Clinical study (glioblastoma RCT), not preclinical. The MGMT-methylation-as-TMZ-predictor mechanism is the relevant import, but Flego 2024 (kept) extends the predictor logic into dSDH-GIST specifically. Hegi 2005 is the foundational clinical anchor and is cited in the validation plan for the MGMT methylation order; including it as preclinical would mis-categorize a phase III trial.

- **Intervention:** MGMT methylation as TMZ-response predictor (Hegi 2005)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Clinical study (glioblastoma RCT), not preclinical. The MGMT-methylation-as-TMZ-predictor mechanism is the relevant import, but Flego 2024 (kept) extends the predictor logic into dSDH-GIST specifically. Hegi 2005 is the foundational clinical anchor and is cited in the validation plan for the MGMT methylation order; including it as preclinical would mis-categorize a phase III trial.